

UV, Optical and NIR Study of SN 2023vbg with a Long Duration Precursor

KAGOSHIMA UNIVERSITY

M1 SOTA GOTO

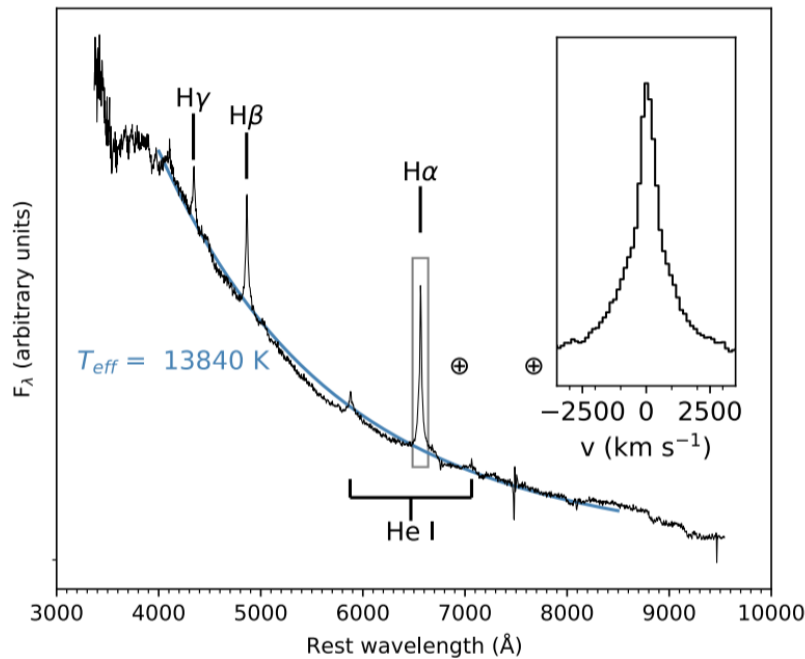
Introduction

The characteristics of Type IIn SN

- Early Spectra :

Blue continuum + Narrow H emission =

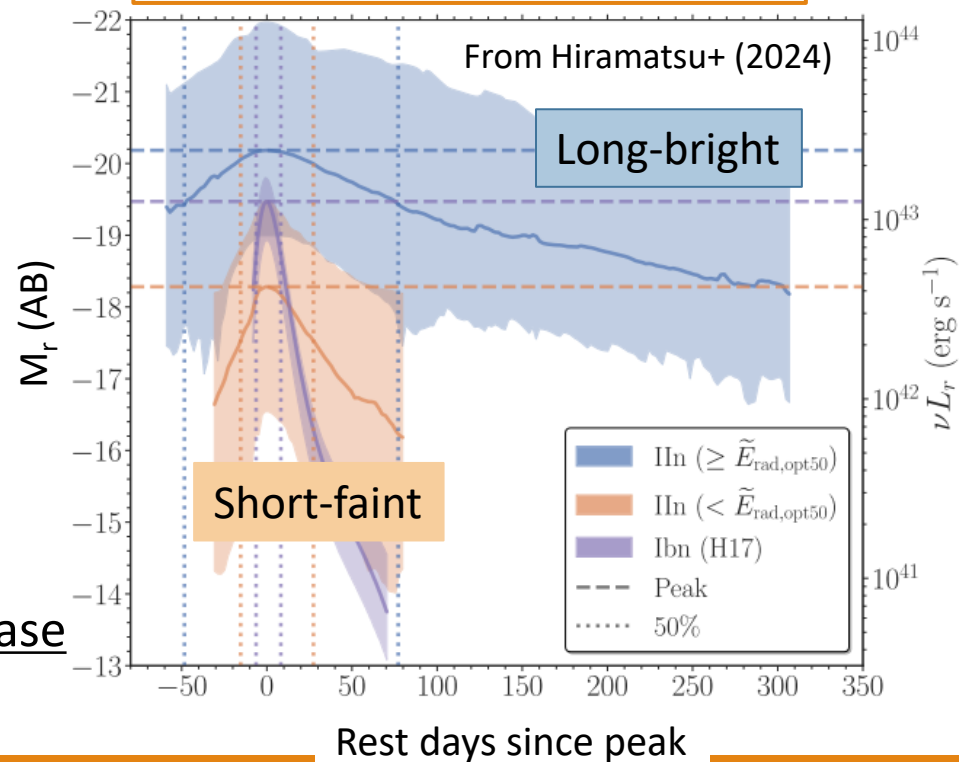
Interaction between
SN ejecta and CSM



From Moran+ (2022)

- H emission are also observed in late phase

Observational Diversity



Introduction

Some SN IIn show “precursor”
⇒ large eruption before SN event

< Well-observed examples >

SN 2009ip (e.g., Mauerhan+ 2013)

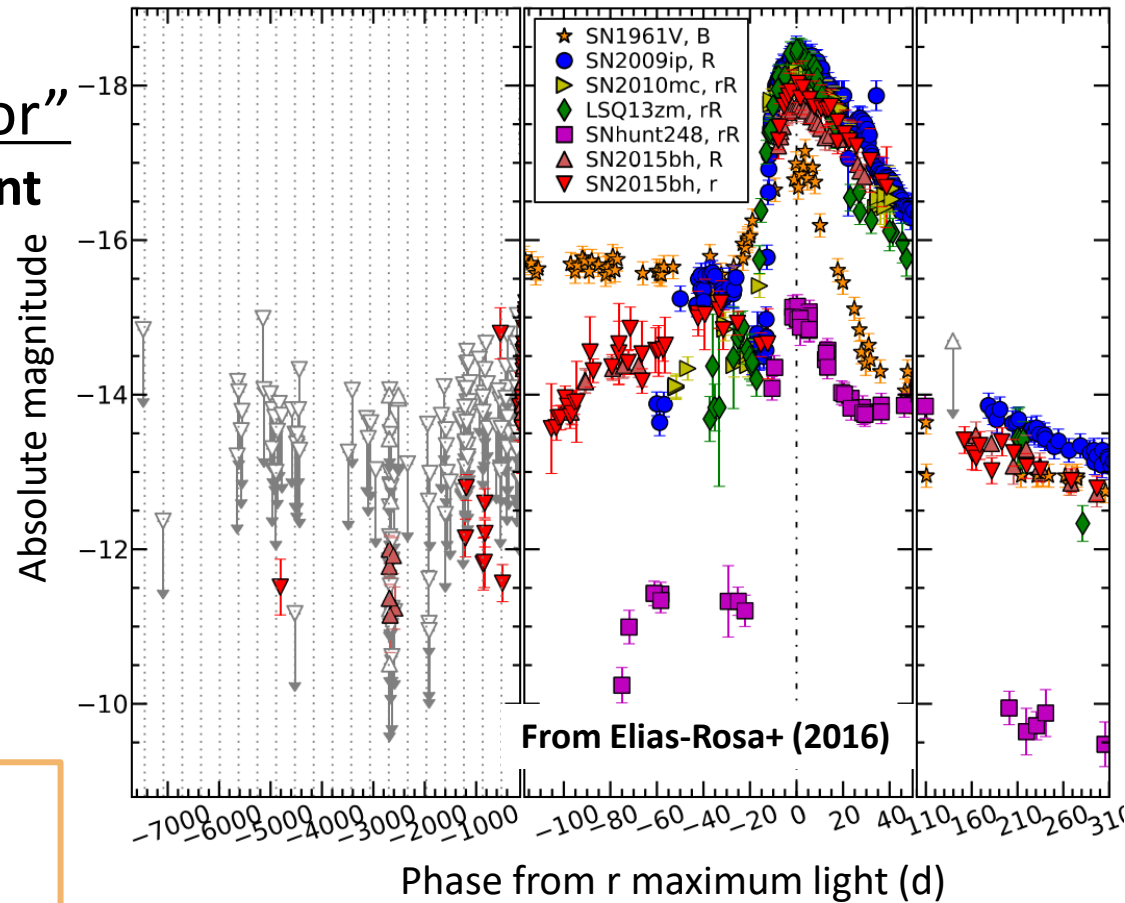
SN 2015bh

(e.g., Elias-Rosa+ 2016)

What is the origin of
precursor activity ??

LBV outburst ??

binary interaction ??



- > Distribution and properties of CSM
- > Mass-loss history before the SN
- > Nature of progenitor

Target of this study

SN 2023vbg (SN IIn)

>Discovery date: 2023-10-15

>Discovery mag (ZTF) : 20.4327 (r)

>Redshift : $z = 0.0173$
(DM = 34.2931)

>Host galaxy :

SDSS J074343.73+342230.2

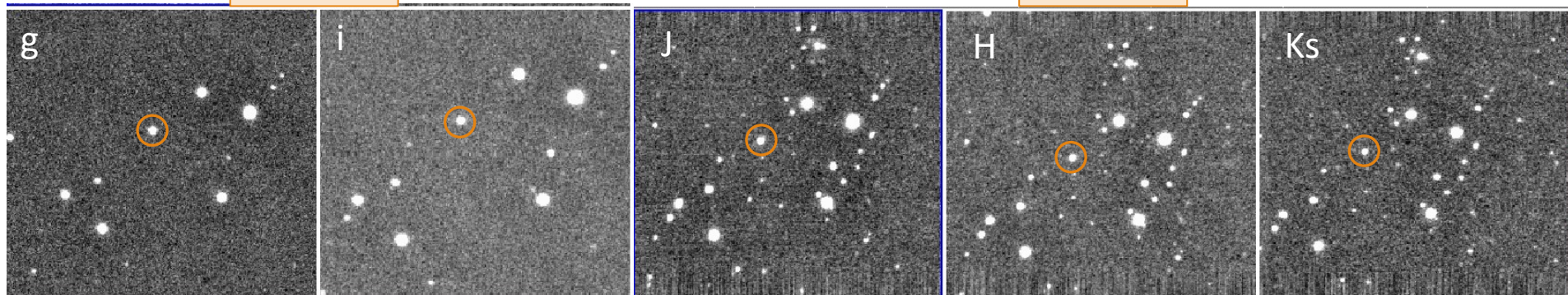
> $E(B-V)_{MW} = 0.05$

Detections of precursor $\Rightarrow$ rapid rise + “**bump**” in rise

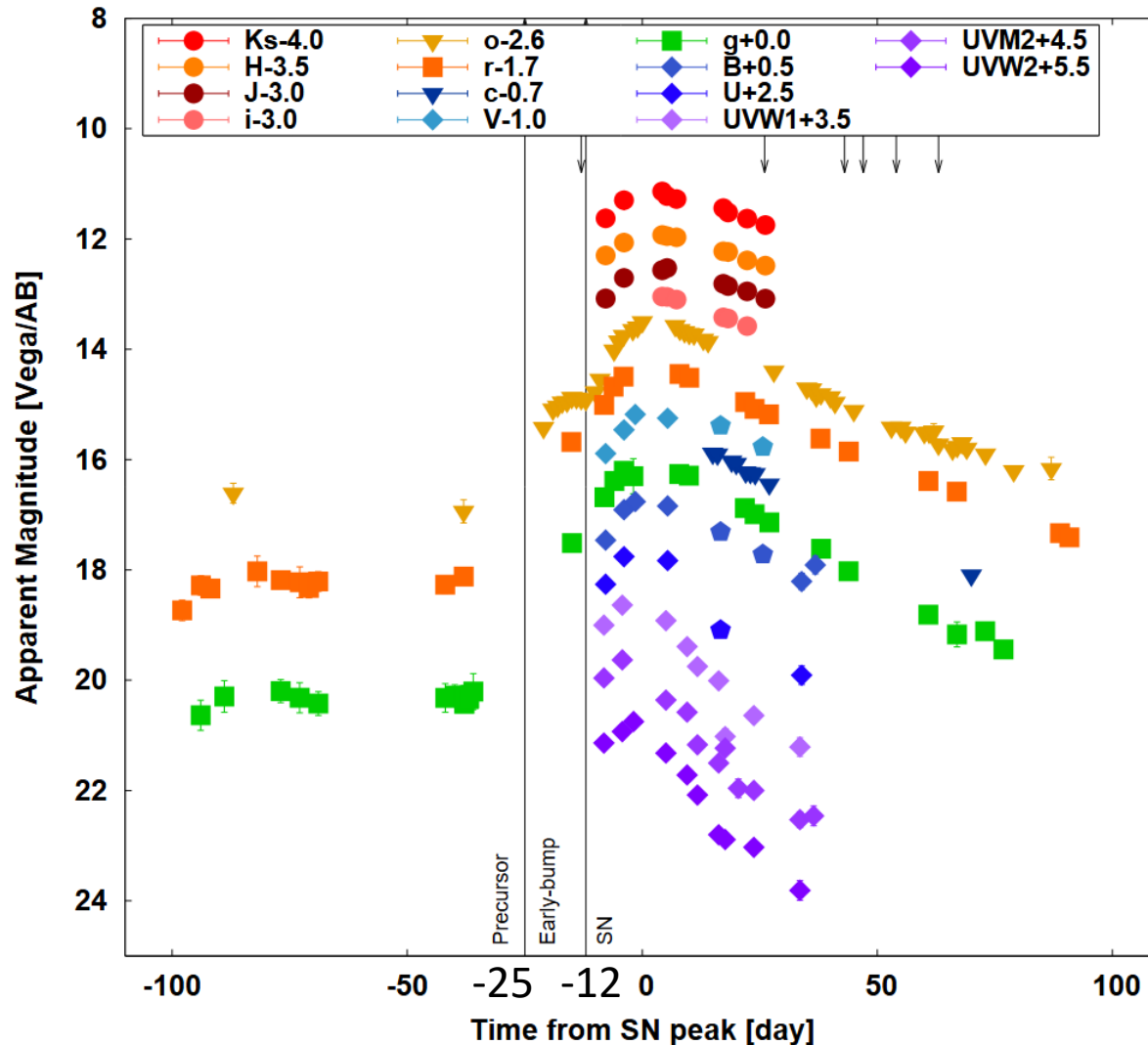
$\Rightarrow$ We started optical + NIR observations this peculiar SN

optcam

kSIRIUS



Multi-band light curve



- $t=0$ is o-band peak of the LC
- Extinctions of MW and host galaxy are not corrected.

We divided LC into 3 phases

Precursor : $t \leq -25\text{d}$

Early-bump : $-25\text{d} < t < -12\text{d}$

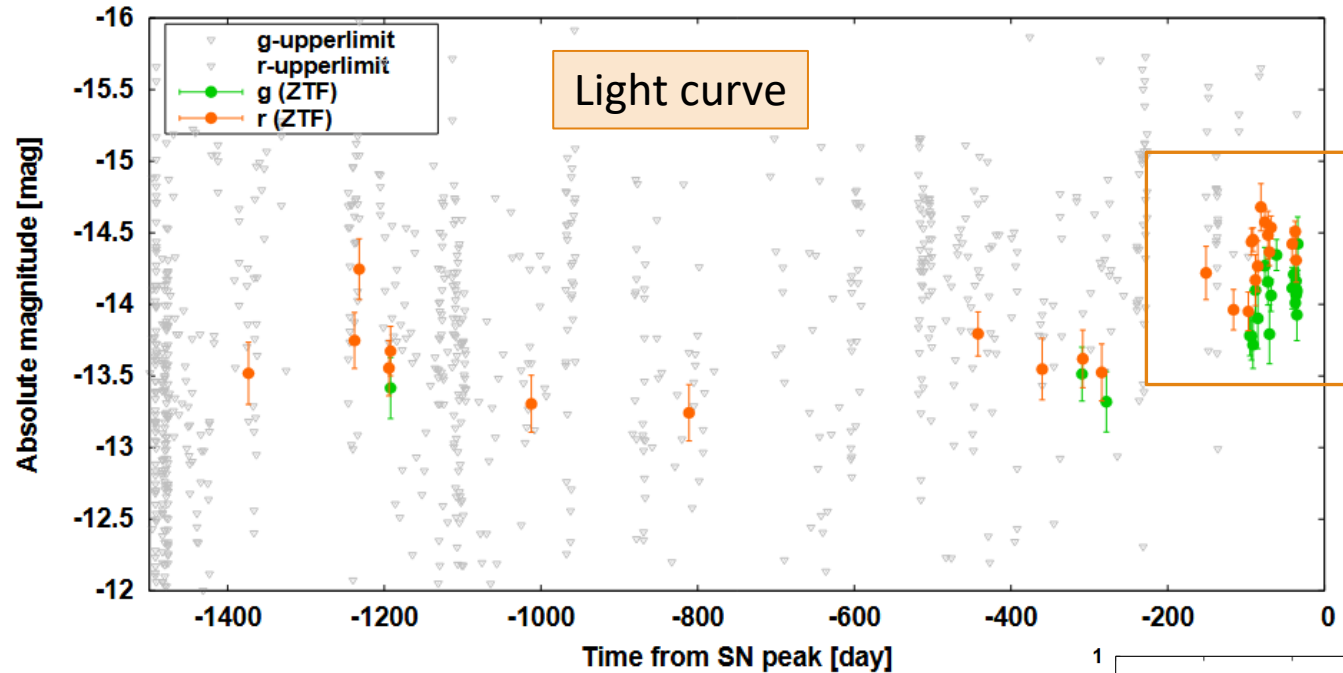
SN event : $t \geq -12\text{d}$



We investigated precursor activity before -100d (using ZTF survey)

Precursor detections

➤ MW extinction is corrected

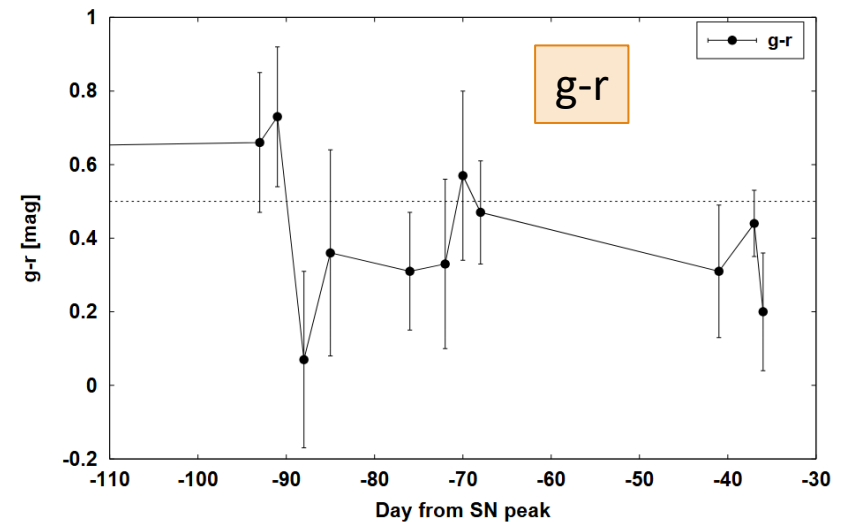


Correspond to precursor before 100days

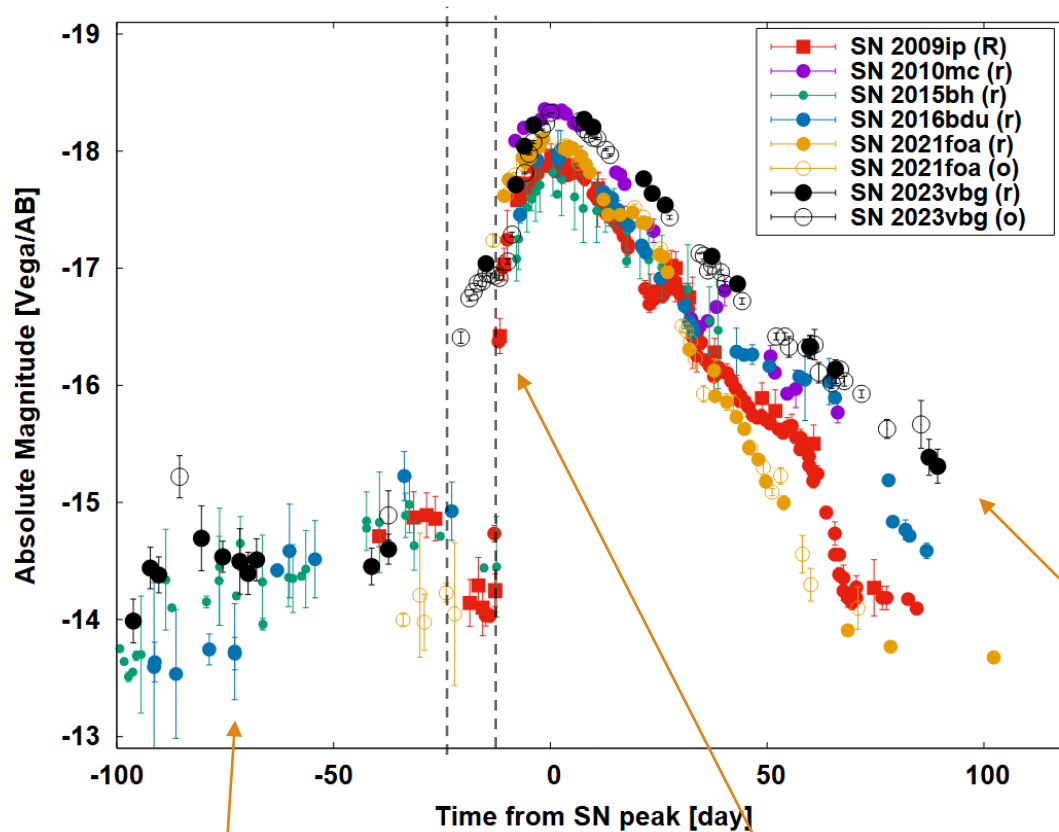
Pre SN outburst g-r : 0.4~0.6mag
(before -25d)



Similar to color in LBV eruption
(Rguitti+ (2024))

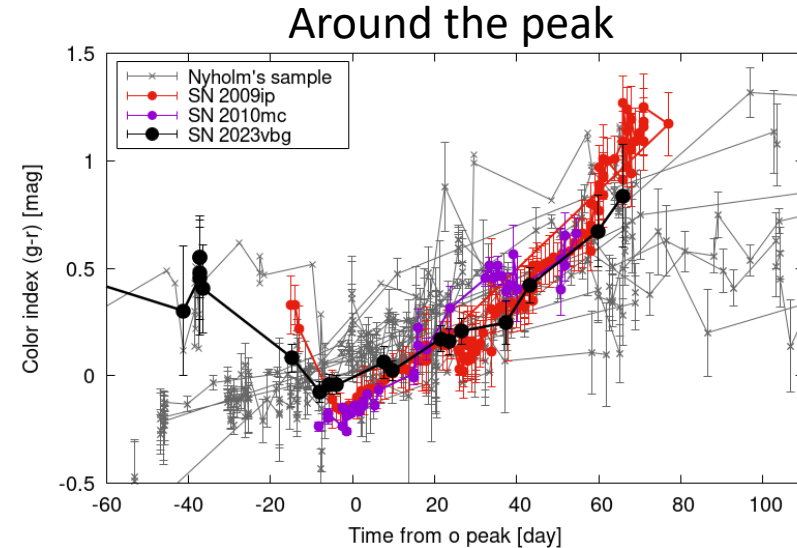


Comparison with other 09ip-like SN IIn



Similar luminosity

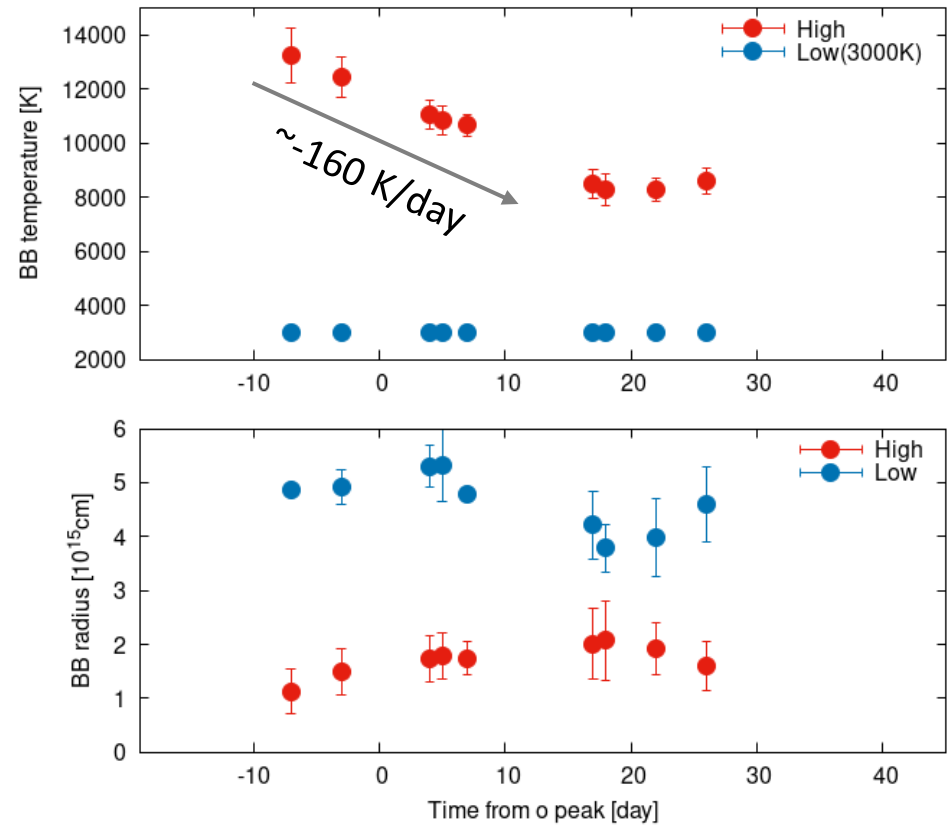
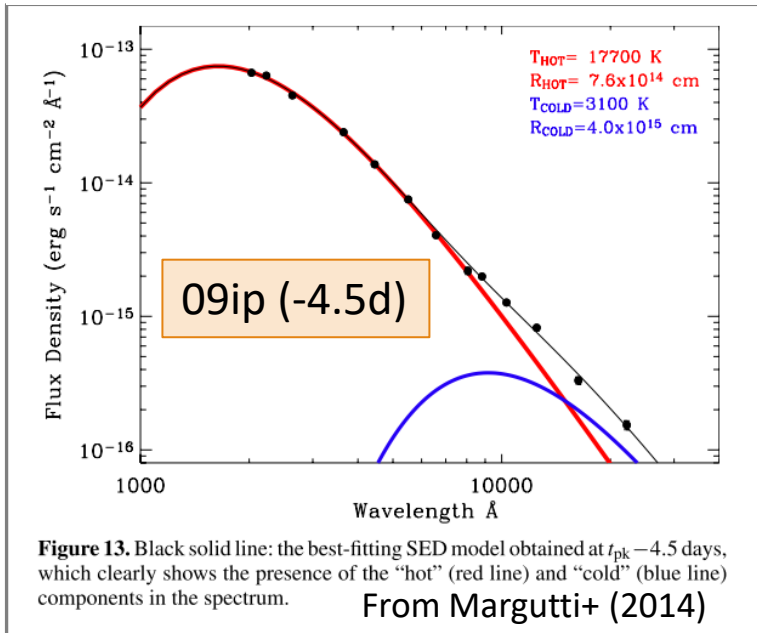
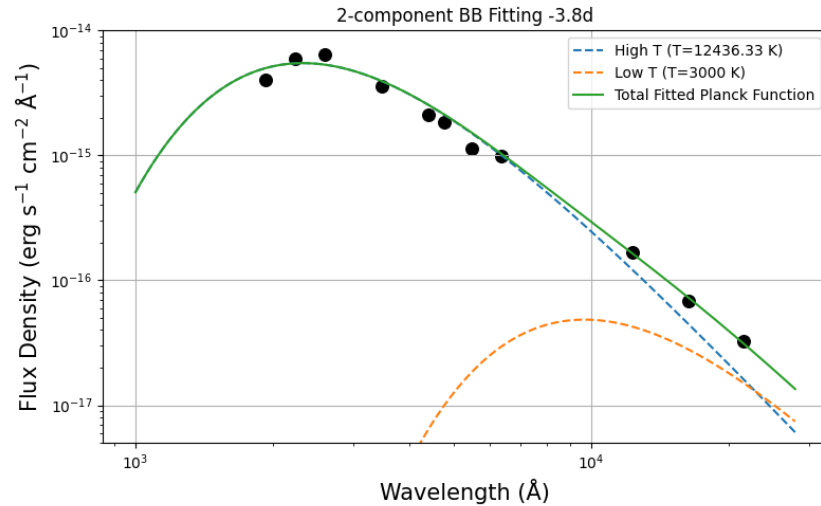
A unique "early-bump" in SN 2023vbg



No significant difference
in color evolution

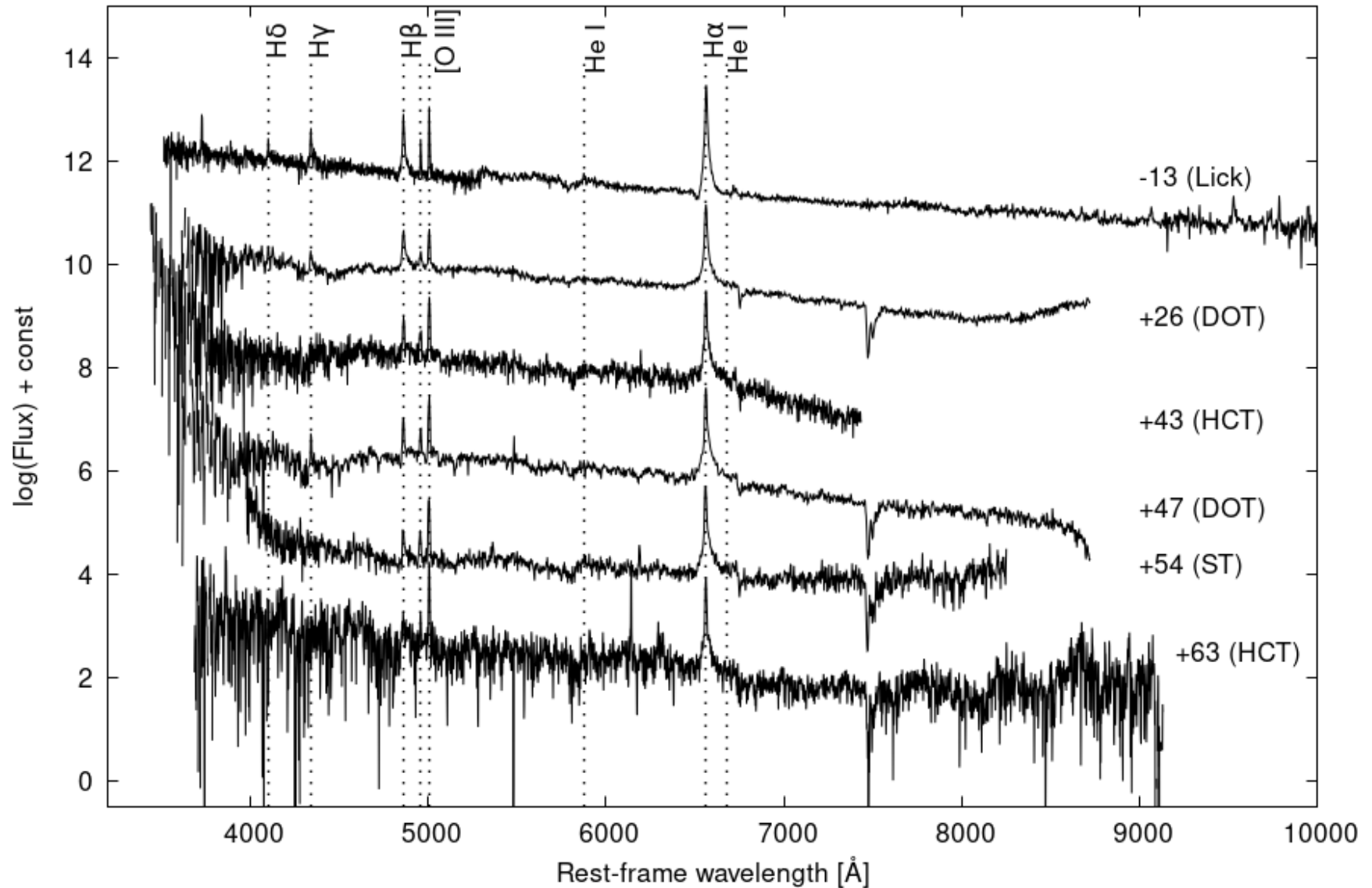
No bump in decline phase

Black body fit

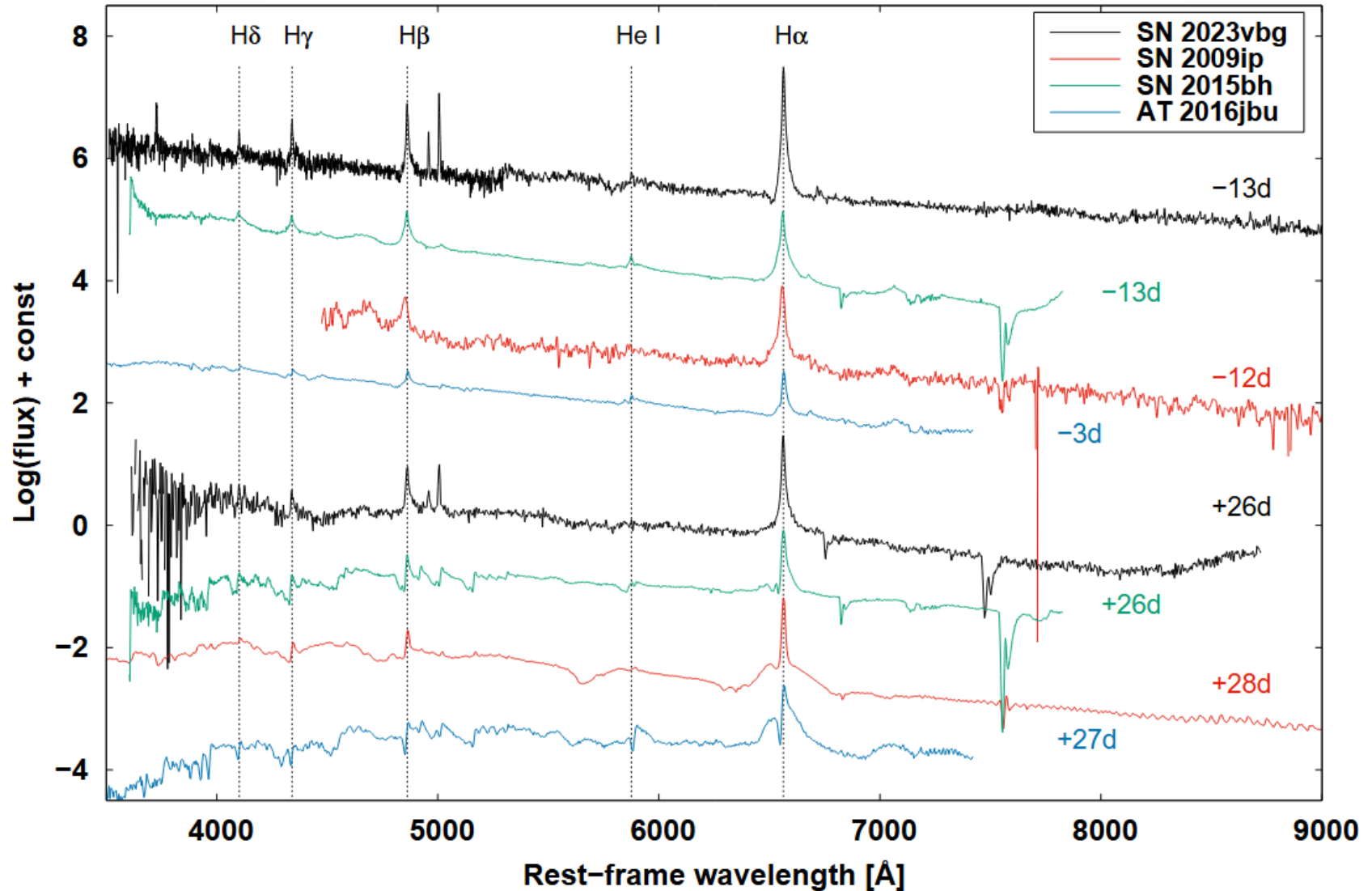


- **NIR excess $\Rightarrow$ two components**
- The effective radius of the "cold" component is greater than twice that of the "hot" one.

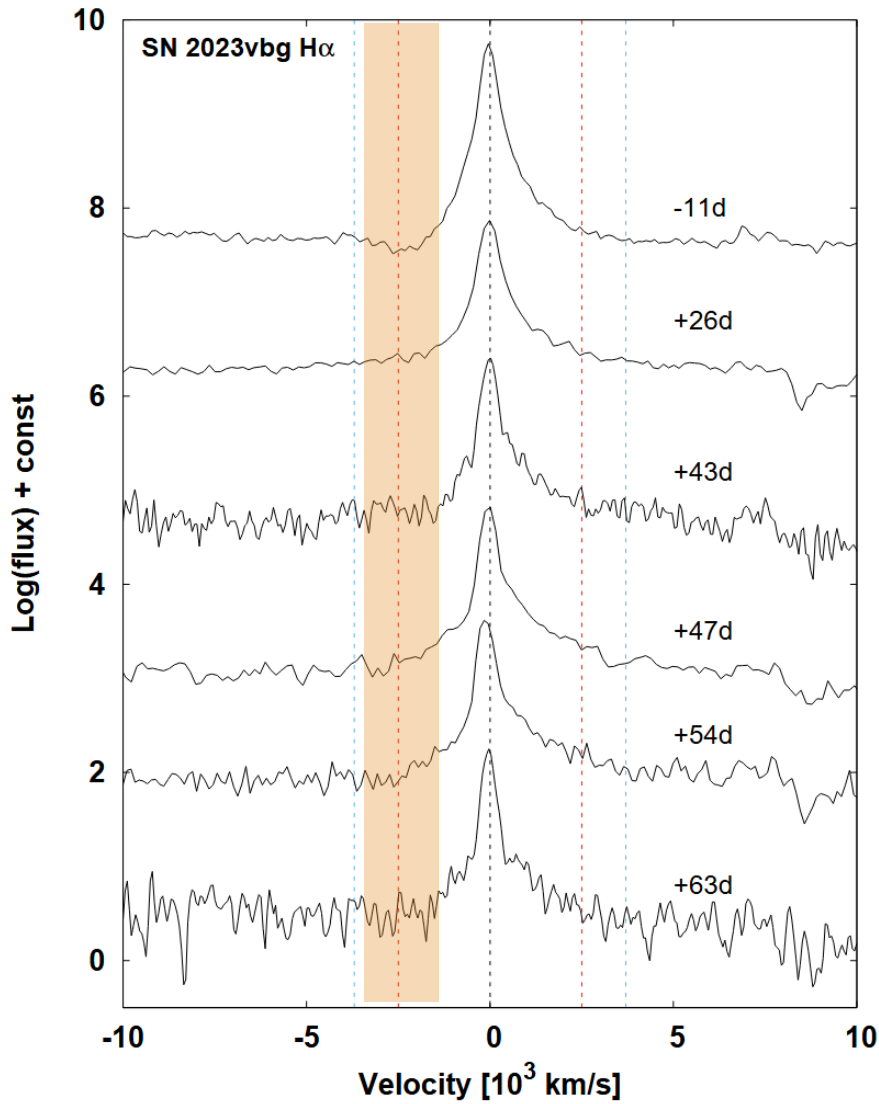
Spectral evolution



Comparison with other 09ip-like SN IIn



H α profile



Blue line = 3500 km/s
Red line = 2500 km/s

H α , H β

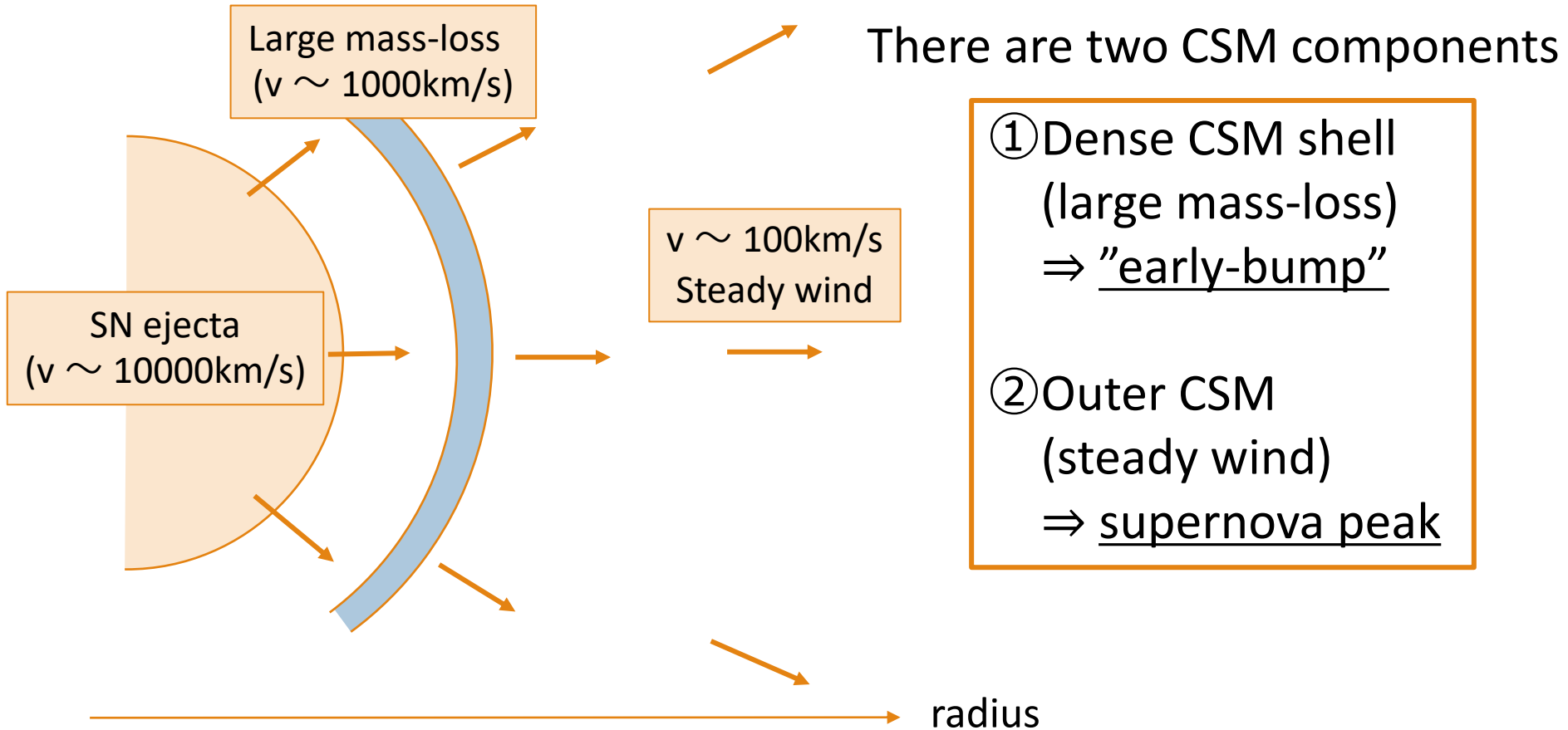
P-Cygni absorption, $v \sim -2500$ km/s
(-13d)



Narrow + intermediate components
(+26d ~)

Discussion

✓ Explosion scenario of 2023vbg



Conclusion

- SN 2023vbg is a 09ip-like Type IIn with **the long-duration precursor event**.
- LC shows a unique "**early bump**" in the rise that is not seen in comparison objects, but **no bump in decline**.
- SED can be explained by the **two-component blackbody** function in the early phase.
- The spectra shows **no significant evolution**.
- Both LC and spectra exhibit **several differences** compared to the other SNe.